

Battery Supply Chain Risk System

LLM Multi-Agent Pipeline · NAATBatt Data (March 2026)

Enter a news article or supply chain query:

Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes.

Sample inputs

S1 — Cobalt mine strike (D... ▾

Load sample

 Analyze Risk

 Analyse abgeschlossen

Severity

 3 / 5

Risk Type

Supply Disrup...

Material

Cobalt

Region

Africa/DRC

Capacity Impact

Affected — NA Upstream

0.0%

Alternative — NA Upstream

100.0%

Affected — Global (USGS)

0.00%

Alternative — Global

2.79%

Top 3 High-Risk Facilities

Rank	Company	Segment	Location	Score
1	Targray Technology Int	Midstream-BGM	QC, Canada	25.2 / 100
2	Targray Technology Int	Midstream-BGM	QC, Canada	25.2 / 100
3	Targray Technology Int	Midstream-BGM	QC, Canada	25.2 / 100



Risk Report

Risk Event

A major strike at Glencore cobalt mines in the Democratic Republic of Congo (DRC) has halted production, affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global cobalt supply will likely persist for at least 1-3 months. Source: NAATBatt facility data + LLM analysis.

Affected Material & Supply Chain Tier

Cobalt, Upstream origin tier, supply_disruption risk type.

Capacity Impact (North America)

- Affected NA capacity: 0.0%
- Alternative NA capacity: 100.0%
- Affected global capacity (USGS basis): 0.0%

Top 3 High-Risk Facilities

1. Targray Technology International Inc. — Montreal, QC, Canada (Midstream-BGM, RiskScore: 25.2/100)
- This facility is high risk due to its significant reliance on DRC-sourced cobalt, which is currently disrupted. Recommended action: Diversify cobalt sourcing to reduce dependence on DRC material.
2. Targray Technology International Inc. — Montreal, QC, Canada (Midstream-BGM, RiskScore: 25.2/100)
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Overall Risk Assessment

Severity 3/5 — A significant disruption in DRC cobalt supply has halted production, affecting 8% of global cobalt output and impacting North American battery manufacturers. The system's resilience is maintained due to alternative NA upstream capacity, which is at 100.0%.

▼ Pipeline details

Validation

 Issues found

2

Iterations

2

Validation issues:

- HALLUCINATION: Glencore cobalt mines mentioned, but Glencore not in the input data
- COHERENCE: Alternative NA capacity is 100.0%, but this contradicts the initial assessment reason stating the strike's impact will persist for at least 1-3 months
- COMPLETENESS: Justification for Targray Technology International Inc. facilities is the same for all three entries
- COMPLETENESS: Recommendation for Targray Technology International Inc. facilities is the same for all three entries

- CONSISTENCY: Severity 3/5 does not match the described event's impact (8% of global cobalt output affected)
- HALLUCINATION: NAATBatt facility data mentioned, but NAATBatt not in the input data

Raw pipeline state (debug)

```
{
  "raw_input" :
  "Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes."
  "iteration" : 2
  "relevant" : true
  "trigger_type" : "A"
  "material" : "cobalt"
  "region" : "Africa/DRC"
  "keywords" : [
    0 : "strike"
    1 : "Glencore"
    2 : "cobalt"
    3 : "Democratic Republic of Congo"
    4 : "wages"
    5 : "prices"
    6 : "production"
    7 : "battery manufacturers"
  ]
  "filtered_text" :
  "Major strike at Glencore cobalt mines in the Democratic Republic of Congo entered its third week Monday, with workers demanding higher wages amid record cobalt prices. The walkout has halted production at facilities supplying roughly 8% of global cobalt output, raising concerns among battery manufacturers in North America who rely on DRC-sourced material for NMC cathodes."
  "severity" : 3
  "risk_type" : "supply_disruption"
  "affected_material" : "cobalt"
  "affected_region" : "Africa/DRC"
  "origin_tier" : "Upstream"
  "reason" :
  "A major strike at Glencore cobalt mines in the DRC has halted production, affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global cobalt supply will likely persist for at least 1-3 months."
  "betroffene_kapazitaet_pct" : 0
  "alternative_kapazitaet_pct" : 100
}
```

```
"risk_report" :
```

```
### Risk Event
```

A major strike at Glencore cobalt mines in the Democratic Republic of Congo (DRC) has halted production, affecting 8% of global cobalt output. This disruption is significant for battery manufacturers relying on DRC-sourced material for NMC cathodes, particularly in North America. The strike's duration and impact on global cobalt supply will likely persist for at least 1-3 months. Source: NAATBatt facility data + LLM analysis.

```
## Affected Material & Supply Chain Tier
```

Cobalt, Upstream origin tier, supply_disruption risk type.

```
## Capacity Impact (North America)
```

- Affected NA capacity: 0.0%
- Alternative NA capacity: 100.0%
- Affected global capacity (USGS basis): 0.0%

```
## Top 3 High-Risk Facilities
```

1. Targray Technology International Inc. – Montreal, QC, Canada (Midstream-BGM, RiskScore: 25.2/100)
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```
## Overall Risk Assessment
```

Severity 3/5 – A significant disruption in DRC cobalt supply has halted production, affecting 8% of global cobalt output and impacting North American battery manufacturers. The system's resilience is maintained due to alternative NA upstream capacity, which is at 100.0%."

```
▼ "top3_facilities" : [
```

```
  ▼ 0 : {
```

```
    "id" : "11008"
    "company" : "Targray Technology International Inc."
    "facility_name" : "nan"
    "segment" : "Midstream-BGM"
    "country" : "Canada"
    "state" : "QC"
    "latitude" : 45.4456782769939
    "longitude" : -73.8750202868265
    "risk_score" : 1.2599999999999998
    "risk_score_normalized" : 25.2
    "tier_weight" : 0.6
    "vulnerability" : 0.7
    "resilience_discount" : 0
```

```
  }
```

```
  ▼ 1 : {
```

```
    "id" : "11009"
    "company" : "Targray Technology International Inc."
    "facility_name" : "nan"
    "segment" : "Midstream-BGM"
    "country" : "Canada"
    "state" : "QC"
    "latitude" : 45.4456782769939
    "longitude" : -73.8750202868265
    "risk_score" : 1.2599999999999998
    "risk_score_normalized" : 25.2
    "tier_weight" : 0.6
    "vulnerability" : 0.7
    "resilience_discount" : 0
```

```
  }
```

```
  2 : {
    "id" : "11011"
    "company" : "Targray Technology International Inc."
    "facility_name" : "nan"
    "segment" : "Midstream-BGM"
    "country" : "Canada"
    "state" : "QC"
    "latitude" : 45.4456782769939
    "longitude" : -73.8750202868265
    "risk_score" : 1.2599999999999998
    "risk_score_normalized" : 25.2
    "tier_weight" : 0.6
    "vulnerability" : 0.7
    "resilience_discount" : 0
  }
]
"global_metrics" : {
  "betreffene_kapazitaet_na_pct" : 0
  "alternative_kapazitaet_na_pct" : 100
  "betreffene_kapazitaet_global_pct" : 0
  "alternative_kapazitaet_global_pct" : 2.79
}
"valid" : false
"failure_type" : "severe"
"issues" : [
  0 : "HALLUCINATION: Glencore cobalt mines mentioned, but Glencore not in the input data"
  1 :
    "COHERENCE: Alternative NA capacity is 100.0%, but this contradicts the initial assessment reason stating
    the strike's impact will persist for at least 1-3 months"
  2 :
    "COMPLETENESS: Justification for Targray Technology International Inc. facilities is the same for all three
    entries"
  3 :
    "COMPLETENESS: Recommendation for Targray Technology International Inc. facilities is the same for all
    three entries"
  4 :
    "CONSISTENCY: Severity 3/5 does not match the described event's impact (8% of global cobalt output
    affected)"
  5 : "HALLUCINATION: NAATBatt facility data mentioned, but NAATBatt not in the input data"
]
}
```